

Advanced Programming Practices

MTech 2026

Lab - 8

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP8_Qn.c (e.g. ABC2026001_APP8_Q1.c).
 - Every program must compile without warnings using the gcc compiler.
 - Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
 - Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
 - Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.
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1. Write a C program to implement polynomial addition using linked lists.
2. Write a C program to create multiple singly linked lists using multiple head pointers. Allow the user to select a particular list and perform insertion, deletion, and traversal on it.
3. Write a C program to maintain three separate singly linked lists using three head nodes. Provide options to insert elements into any selected list and display all three lists.
4. Write a C program to represent a sparse matrix using a 3-tuple array. Display the sparse matrix in triplet form containing row, column, and value.
5. Write a C program to search for an element at a specified row and column in a sparse matrix represented using a linked list.